

Integrated Carbon Capture and Utilization

From Flue Gas to Polymer: Carbon
Capture Integrated with Methanol-to-
Monomer Production

Hiba-Batool Naqvi

Angelica Ramirez

Javier Silva

Cesearo Dilosa

Ify Bidokwu

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1 Project Charter



2 Executive Summary

The project plans for a pathway of carbon utilization that converts post-combustion CO₂ from a steel refinery located in Gary, Indiana to valuable light olefins (ethylene and propylene) using an integrated CO₂ to methanol to olefins process. The system proposes to address both challenges of industrial decarbonization and a growing demand for polymers by changing wasteful gas to a key petrochemical feedstock.

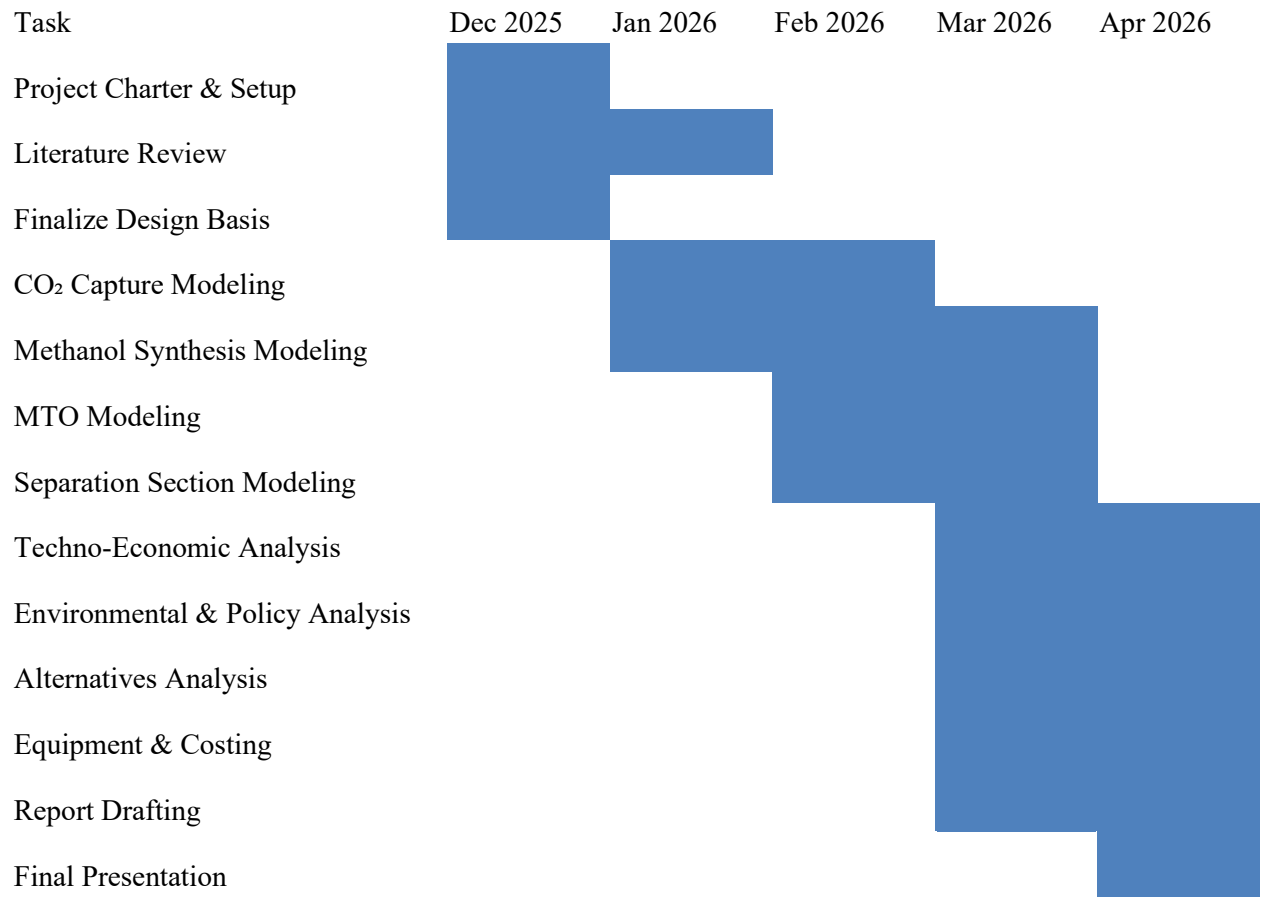
The capture is purified and then reacted with hydrogen over a catalyst of ZnO/Cu/Al₂O₃ to produce a high-purity methanol. This methanol is then converted in a Honeywell-UOP-style methanol-to-olefins (MTO) reactor with the use of a SAPO-34 catalyst system to produce 830,000 tons per year of olefins (469,000 tons per year of ethylene and 361,000 tons per year propylene). The design basis of the project assumes 8000 operating hours per year and 20 years of plant operations.

Considering the economic structure of the project, the process utilizes the value of methanol upgrading and U.S. potential 45q tax credits for CO₂ usage. Environmentally, the project proposes a system that reduces reliance on fossil-based steam cracking and provides sustainable olefin production. Technologically, both MTO and CO₂ to methanol are processes with significant amounts of industrial research and development.

The project explores the MTO configuration and analyzes other alternative methanol-to-polymer routes including methanol-to-polymer (MTP), methanol-to-gas (MTG) with oxidating coupling variants, olefin recovery, and direct methanol-to-polyolefin ideas. These alternative processes are seen as comparisons for long term economic strength and scalability.

The final design demonstrates that CO₂ methanol is a feasible platform chemical for sustainable polymer production, with the provision of a hydrogen source and optimized catalyst management. The integrated system places carbon capture as a valuable chemical manufacturing method.

3 Gantt Chart



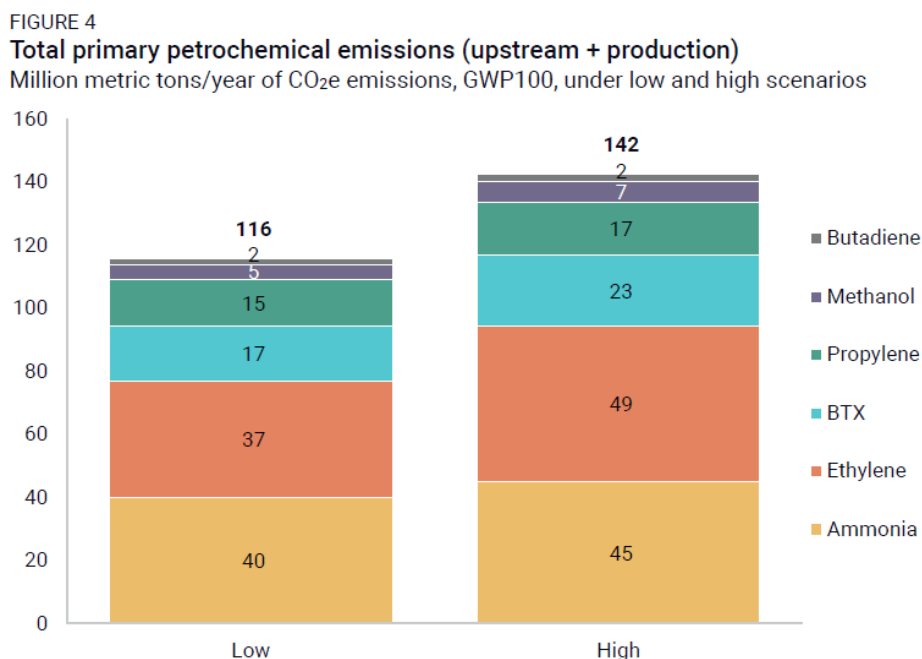
4 Introduction

4.1 Problem Definition and Significance

Global demand for light olefins, primarily ethylene and propylene, is steadily increasing due to their crucial role as precursor building blocks in the production of polymers, fibers, chemicals, and consumer materials. The International Energy Agency (IEA) reports that olefin consumption continues to rise at ~3–4% annually, making petrochemicals one of the fastest-growing sources of oil demand worldwide.

Traditionally, ethylene and propylene are produced through steam cracking of hydrocarbons derived from crude oil and natural gas. This conventional method is highly energy-intensive, environmentally damaging, and responsible for substantial CO₂ emissions and industrial waste. For example, ethylene production alone releases approximately 1.5–3.0 tonnes of CO₂ per tonne of product, depending on feedstock and energy source.

These fossil-based processes contribute heavily to global greenhouse gas emissions, resource depletion, and worsening climate impacts. The chemical industry now accounts for ~18% of global industrial CO₂ emissions, with light olefin production being a major contributor.



Source: Rhodium Group, Global Data.

Notes: BTX is a summation of the emissions from benzene, toluene, and xylenes.

Figure 1: Total Primary Petrochemical Emissions (Bower 2025)

To address this issue, there is a growing focus on closing the carbon loop. Rather than releasing CO₂ during olefin production, captured CO₂ can be utilized as the carbon feedstock, providing a green alternative that replaces petrochemical extraction and mitigates emissions from existing industrial activity.

4.2 Carbon Capture Motivation

Carbon Capture, Utilization, and Storage (CCUS) is recognized as a critical climate mitigation pathway by global scientific and policy organizations. Industrial flue gas, especially from energy-intensive manufacturing such as steelmaking, contains large volumes of CO₂ with concentrations that make capture economically viable. Steel manufacturing alone contributes ~7–9% of global CO₂ emissions (Hanson 2025).

Gary, Indiana is home to one of the largest integrated steel plants in the U.S., producing millions of tonnes of flue-gas CO₂ annually. This provides a convenient and continuous source of carbon for upgrading into valuable chemical products.

By integrating carbon capture with chemical conversion technologies such as CO₂ hydrogenation and methanol-to-olefins (MTO) synthesis, previously emitted carbon can now be preserved within a circular process. This reduces direct emissions from steel operations, dependence on fossil hydrocarbons, net global atmospheric carbon intensity

The approach not only prevents waste emissions but transforms CO₂ into a revenue-generating feedstock, aligning environmental sustainability with industrial profitability.

4.3 Importance of Sustainable Polymer Production

The shift toward sustainable polymer manufacturing is a major transformative objective across chemical and consumer product industries. Traditional plastics rely on finite fossil resources and create long-lived waste streams. By contrast, sustainable polymer frameworks prioritize Carbon-neutral feedstocks, advanced catalysis to lower energy consumption and improve selectivity, circular-economy recycling methods, and low-emission energy integration across chemical plants

Organizations such as the National Academies and IEA note that integrating CO₂ utilization into chemical value chains could reduce global carbon emissions by billions of tonnes annually while supporting economic growth.

Sustainable polymer production, lowers GHG emissions across the entire product lifecycle, reduces reliance on crude-oil-derived hydrocarbons, encourages innovation in lightweight, high-performance materials, and supports global decarbonization legislation and ESG goals

By converting CO₂ into monomers such as ethylene and propylene, materials already fully compatible with existing polymer infrastructure, this approach accelerates the transition toward a carbon-circular plastics economy without requiring redesign of consumer products.

5 Market Analysis

5.1 Global Polymer Market

The global polymer industry is one of the most significant segments of the chemical value chain, encompassing the packaging, construction, automotive, textile, and consumer goods. Packaging remains the most significant industry for both polyethylene (PE) and polypropylene (PP). Recent market estimates project the global market value to be USD 790-800 billion in 2024, with a forecast to reach USD ~1.34 trillion by 2034. Corresponding to a compound annual growth rate (CAGR) of 5-5.5% over the next decade (Research, Polymers Market Strengthen industrial application with bio-based materials, lightweight thermoplastics, and Asia-Pacific scale 2025). Specifically, thermoplastics account for more than 40% of this value, with polyolefins, such as polyethylene and polypropylene, representing the dominant volume share.

Polyolefins are the primary downstream product of ethylene and propylene, and therefore the most relevant to this process. The global polyolefin market size is estimated at USD 262.71 billion in 2024 and projected to reach USD 357.39 billion by 2030, growing at a CAGR of 5.4% from 2025 to 2030 (Region 2024).

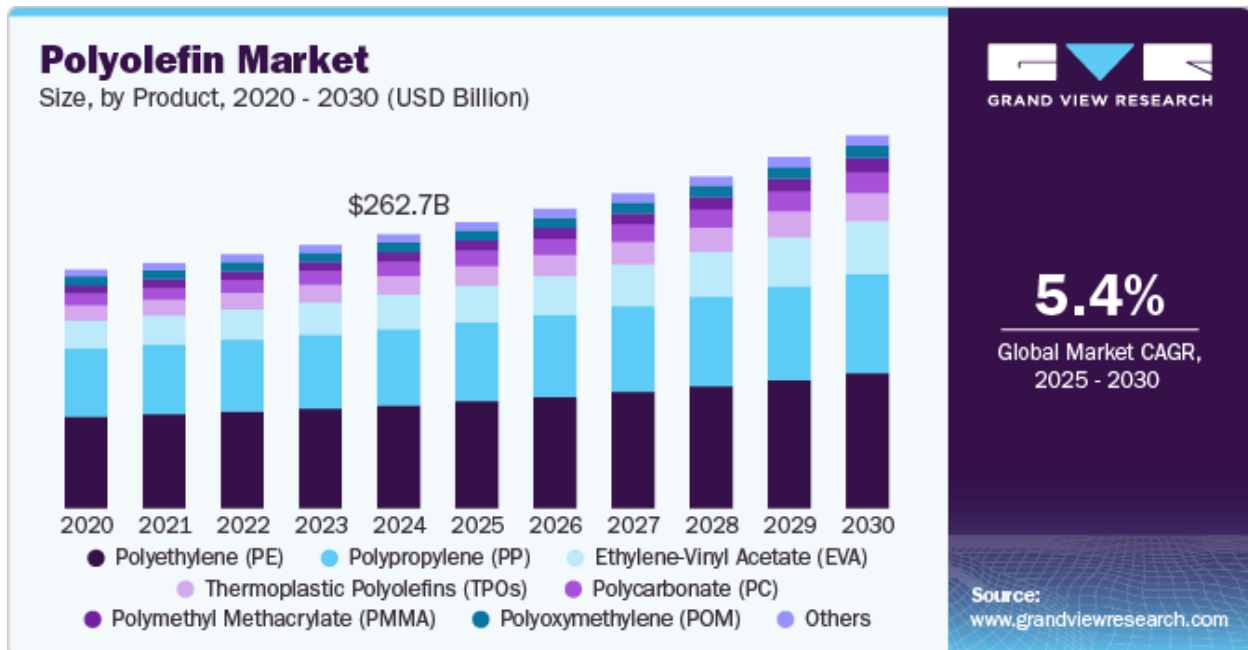


Figure 2: Polyolefin Market (Region 2024)

At an individual polymer level, the PE market was valued at around USD 150 billion in 2022 and is expected to reach 190 billion in 2030, growing at a CAGR of 2.8% from 2023 to 2030 (End-use 2024). PP is estimated to be USD 123 billion in 2022 and expected to reach USD 177.73 billion in 2030, with a DAGR of 4.7% from 2023 to 2030 (Process 2024).

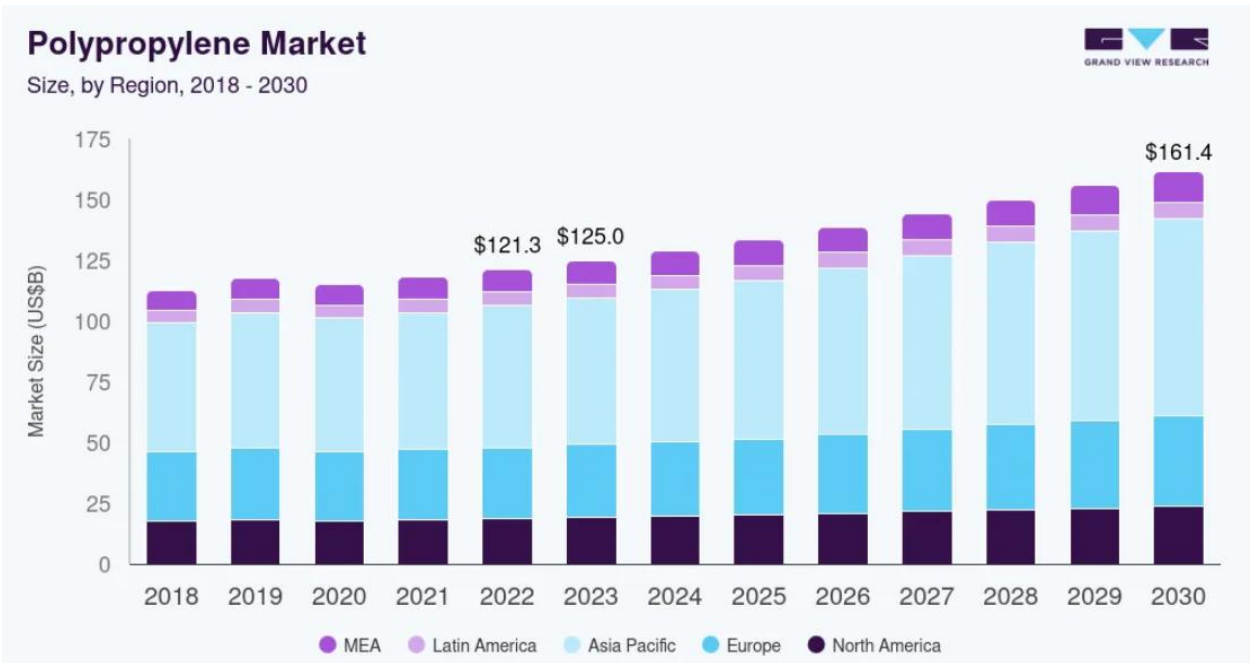


Figure 4: Polypropylene Market (Process 2024)

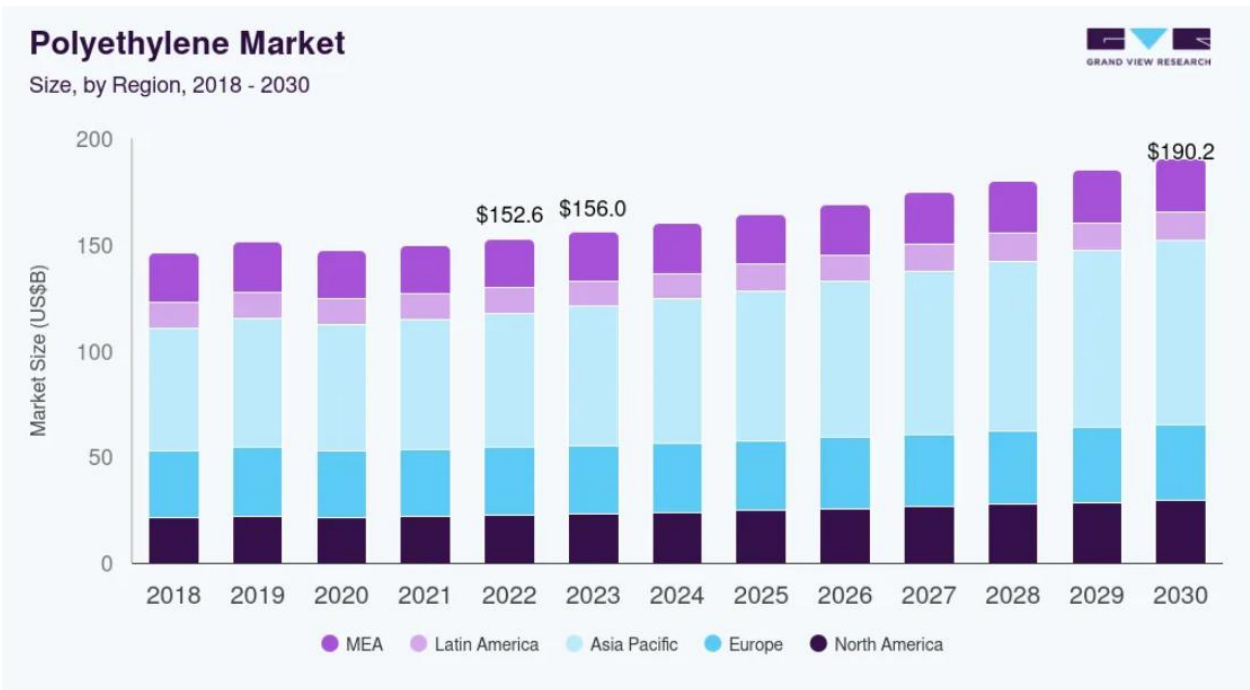


Figure 3: Polyethylene Market (End-use 2024)

In terms of region, the Asia Pacific dominates the market, accounting for 49.95% of the market share in 2024. Specifically, China's olefin market is expected to be driven by the domestic demand, expansion of the production capacity, and the material's diverse applications. In contrast, polymer demand in Europe is relatively mature and currently constrained by high energy costs and regional restrictions imposed by petrochemical companies. North America shows moderate growth, supported by low-cost ethane feedstock and export-oriented investments in PE and PP capacities.

Polymer Markets remain cyclical, with profitability heavily influenced by crude oil, natural gas, and NGL feedstock prices, as well as over-capacity cycles in steam cracking and PDH units. Recent analyses have highlighted an oversupply situation since 2022, which has been pressuring margins and prompting plant closures or rationalization in higher-cost regions. Nevertheless, long-term projections still indicate positive demand growth for both commodity and engineering plastics, with value shifting toward specialty-grade, high-performance materials and applications that replace heavier or more energy-intensive materials.

A significant trend is the transition towards more sustainable polymers and circular plastics. Markets for "sustainable" or circular plastics, including recycled, bio-based, and CO₂-derived polymers, are expected to grow significantly faster than conventional plastics, with a predicted CGR value of >9-13% over the next decade (Research, Commodity Plastics Market Size to Worth USD 666.76 Billion by 2034 2025). Specific drivers for this market's growth include the EU's mandate for recycled content targets in packaging, producer responsibility schemes, bans and taxes on single-use plastics, and corporate net-zero commitments from major FMCG, automotive, and retail companies. These trends do not eliminate the need for ethylene and propylene in the near future, but instead shift the preferred source and carbon footprint of the monomers. Technologies that enable low-carbon or circular feedstock routes are therefore increasingly attractive as part of decarbonization strategies for polymer value chains.

Within this context, an 830 ktma plant represents a small volume in the global polyolefin market, but it is significant on a regional scale. Furthermore, the proposed plant addresses both the growing demand for polyolefins and the pressure to reduce life-cycle greenhouse-gas emissions associated with conventional monomer production. Overall, the global polymer market is large and growing, and as a result, it is becoming increasingly carbon-constrained. This drives the need for a more robust foundation and strategic route for developing a CO₂-based methanol-to-olefin process, which will provide a lower-carbon source of polymers.

5.2 Value Chain Overview

The value chain for this project spans the transformation of captured carbon dioxide into a high-value polymer. This pathway leverages clean energy, catalytic conversion technologies, and established downstream polymer markets to create both economic and environmental value. The proposed plan begins with CO₂ as a feedstock, proceeds through methanol synthesis, and then to a methanol-to-olefin (MTO) conversion, ultimately producing light olefins. At each stage, value is added, but costs and carbon-intensity risks also accumulate. More importantly, the ultimate competitiveness and sustainability of the chain depend on two key drivers: the cost and carbon footprint of the hydrogen used, as well as the availability of policy incentives.

CO₂, captured from steel flue gas, is purified and fed into a hydrogenation loop where H₂ and CO₂ react to produce methanol. The hydrogen is supplied either externally or generated on-site through electrolysis or reforming with carbon capture. This methanol serves as an intermediate, a liquid, transportable, and storable chemical that can substitute for conventional fossil-derived methanol or feedstocks. From methanol, the core MTO reactor converts the methanol into light olefins (ethylene, propylene) plus some byproducts. These olefins serve as the foundational building blocks for polyolefin elastomers, solvents, and numerous petrochemical derivatives. Thus, the chain bridges waste or excess CO₂ to high-value commodity chemicals, effectively recycling carbon. The marketing of this product will focus on this aspect, as well as its strong sustainability positioning. By leveraging CO₂ utilization and a lower-carbon footprint, buyer demands for environmentally improved materials are satisfied. Certification, emissions disclosure, and compliance with circular economy targets become strategic market differentiators. Customers are incentivized to adopt these materials to meet corporate ESG goals and comply with regulatory mandates regarding the use of recycled or low-carbon plastics.

However, the value of this chain is susceptible to the hydrogen source and its carbon intensity. Suppose hydrogen is produced by conventional steam-methane reforming without effectively producing grey hydrogen. In that case, the upstream CO₂ reduction may be negated by the CO₂ emitted during hydrogen production, undermining any climate advantage and increasing the net carbon footprint. Suppose hydrogen is supplied via blue hydrogen or green hydrogen. In that case, the carbon intensity of the overall olefin output drops significantly, and the overall value of the process becomes negative, consequently resulting in a loss of money due to the high cost of these types of hydrogen. The most significant solution to this would be to produce the grey hydrogen in plant, while simultaneously capturing the carbon. However, the logistics of this need to be researched to implement. The decision of whether to purchase hydrogen or integrate on-site hydrogen production further influences operating costs and plant autonomy.

The economics of this chain are also heavily dependent on the government and regulatory incentives around carbon capture and utilization. In a U.S. context, the tax-credit framework under Section 45Q offers financial incentives per metric ton of CO₂ captured and either stored or utilized (Jones 2023). Under Section 45Q, a facility capturing CO₂ and utilizing it in chemical conversion may qualify for the utilization credit rate (IRS 2025). Section 45Q tax credits provide up to \$60 per metric ton of CO₂ utilized and up to \$130 per metric ton stored permanently, depending on the pathway and compliance requirements, according to congressional analysis. This incentive may significantly improve the economics of upstream carbon capture, provided the facility meets the eligibility criteria.

A crucial consideration in this value chain is that monomers alone do not maximize profitability. Ethylene and propylene are globally traded commodities with volatile margins and are most economically valuable when converted into higher-value derivatives, such as polymers. Polyethylene and polypropylene command significantly higher selling prices and more stable demand profiles than their monomer precursors; therefore, a strategic integration of downstream polymerization units is necessary to capture the value of this conversion process fully. Without access to polymerization, the project would remain exposed to monomer price cyclicalities and face reduced returns on investment.

Given these dependencies, the value of this project hinges on the costs of hydrogen sourcing, carbon intensity, final product form, and the policy environment. With green or blue hydrogen and a favorable tax or credit regime, such as 45Q, CO₂-derived olefins can become both economically competitive and environmentally attractive. Without them, the process may struggle to compete with conventional, well-optimized steam-cracking-based olefin production, especially in regions with low-cost fossil feedstocks. This value chain demonstrates how the transformation of low-value CO₂ into polymer materials represents a complete industrial upgrade pathway. Each step adds value, from waste carbon to commodity chemicals to multipurpose manufactured materials, while also aligning with policy and regulatory incentives that support decarbonization. Achieving the greatest value capture requires integrated polymerization capacity, rather than selling monomers alone, ensuring that the economic benefits of the conversion process are fully realized at the highest-value point in the product chain.

5.3 Pricing of Raw Materials

The selection, cost, and sourcing strategy of raw materials represent critical drivers of both the economic and environmental performance of the proposed CO₂-to-polymers facility. The major inputs to this process include captured carbon from flue gas, hydrogen, water, and supporting chemicals.

Methanol is both a potential feedstock and a reference commodity. As of late 2025, contract methanol price indicates a typical methanol value around USD 320 per tonne in

the North American region, and USD 250-290 per tonne in the Asian Middle Eastern Region (Price-Watch 2025). Given these market numbers, the raw-material cost baseline is approximately USD 300–350/tonne, depending on the region, freight, and purity requirements. This baseline helps contextualize the competitiveness of methanol produced from CO₂ conversion within internal operations.

Hydrogen is a critical raw material that directly influences the cost and carbon footprint of the entire chain. Current hydrogen prices depend heavily on production method: grey, blue, and green. Grey hydrogen is from natural gases and carbon capture. It remains the most cost-effective option, with recent market prices ranging from USD 0.96 to USD 3 per kilogram. Blue hydrogen is a fossil-driven form of hydrogen that also utilizes carbon capture. It has a higher cost, typically around USD 2-3.5 per kilogram, reflecting the purification costs. Ultimately, green hydrogen is produced through water electrolysis using renewable energy sources. It is currently the most expensive, with typical market values ranging from USD 3.5 to USD 6.0 per kilogram. Due to the high hydrogen demand per tonne of methanol for the CO₂ to methanol process, the choice of hydrogen supply method becomes a major lever in the project. From this, the scenario of using grey hydrogen in combination with a circular CO₂ capture method becomes the most profitable.

Flue gas is not purchased as a raw material but instead can act as a potential profit source. As previously discussed, tax credits are available for up to \$60 per metric ton of CO₂ utilized. It is crucial in this process to be accredited by the government and to be subject to its regulation to take advantage of this potential source of profit.

Water is required for hydrogen production, cooling, and diluent/reactant in various units. In many industrial contexts, the cost of raw water is relatively low compared to the costs of fuel and hydrogen; however, in some geographies, water scarcity may make water a non-trivial operating expense. For rough economic modeling, water can often be assumed at a modest tariff or treated as an internal utility cost. Utility costs (electricity for H₂, steam, cooling water, and compression) should be treated separately in the energy balance and utility section, rather than as raw materials per se.

Other chemicals, such as catalysts, also contribute to operating costs but are best handled separately under reagents and operating consumables rather than raw material costs for bulk inputs.

<i>Table 1: Raw Materials Pricing Table</i>			
Raw Material	Pricing Basis	Price Range	Notes
Grey Hydrogen	USD/kg	0.96-3.0	Low cost but highest carbon intensity

Blue Hydrogen	USD/kg	2.0-3.5	Includes cost of carbon capture and removal infrastructure
Green Hydrogen	USD/kg	3.5-6.0	Cleanest option
Flue Gas	-	-	Offset by tax credits
Water	USD/m ³	1.00	Specific to Gary, IN
Amine Solvent	USD/kg	1.0-2.0	Consumed gradually over operating life
SAPO-34 Catalyst	USD/kg	10-50	Periodic replacement
Cu/ZnO/Al ₂ O ₃ Catalyst	USD/kg	20-80	Periodic replacement

5.4 Pricing of Products

The economic viability of the MTO-based plant depends not just on the raw material costs but also on the market value of the output products. This process produces two main products, ethylene and propylene, as well as numerous byproducts that can also be sold for profit.

As of late 2025, the cost of ethylene in the NA region is USD 0.56-0.6 per kilogram. Propylene prices are assessed to be USD 0.83-0.84 per kilogram. In contrast, the prices of the polymerized monomers, polyethylene and polypropylene, are roughly USD 1.05-1.2 per kilogram. This is significantly higher than typical monomer prices and reflects the need for the added polymerization step within the process. The jump from monomer to resin adds considerable margin, provides direct access to high-demand resin markets, and is likely essential to realize the full economic potential of the plant.

<i>Table 2: Pricing of Products Table</i>			
Product	Pricing Basis	Price Range	Notes
Ethylene	USD/tonne	560-600	NA 2025 price
Propylene	USD/tonne	830-840	NA 2025 price
Polyethylene	USD/tonne	1110-1200	For a general-purpose resin grade
Polypropylene	USD/tonne	1050-1120	For a general-purpose resin grade
Fuel Gas (Methane)	USD/tonne	120-160	
Ethane	USD/tonne	150-220	Assuming low grade

C ₃ ⁺ Heavier Hydrocarbons	USD/tonne	220	Sold as mixed NGL/fuel-range stream
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6 Judging Criteria

Due to rising global CO₂ emissions, up 0.8% in 2024, reaching an all-time high of 37.8 Gt CO₂, there is a strong incentive to develop alternative and more sustainable industrial processes. Much of this increase stems from industrial activity, which has contributed to a 50% rise in atmospheric CO₂ concentrations (ppm) since 1750.

Currently, the combustion of natural gas, liquid fuels, and coal remains the primary driver of long-term carbon emission growth. Countries such as China, Japan, Indonesia, India, and the United States are among the largest contributors due to their substantial energy consumption. However, it is worth noting that the United States and Japan have seen a steady decline in CO₂ emissions since the early 2000s.

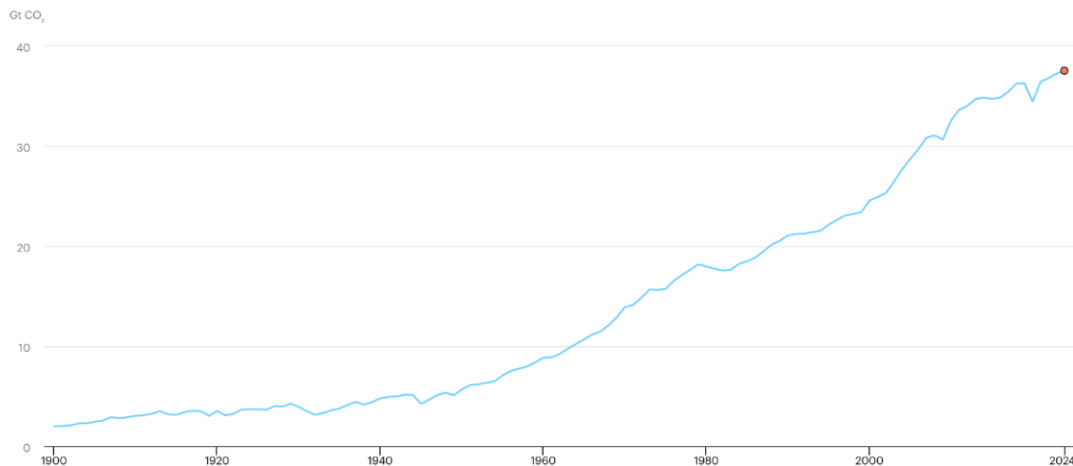


Figure 5. Global CO₂ Emissions Trend (1900-2024), (IEA 2025).

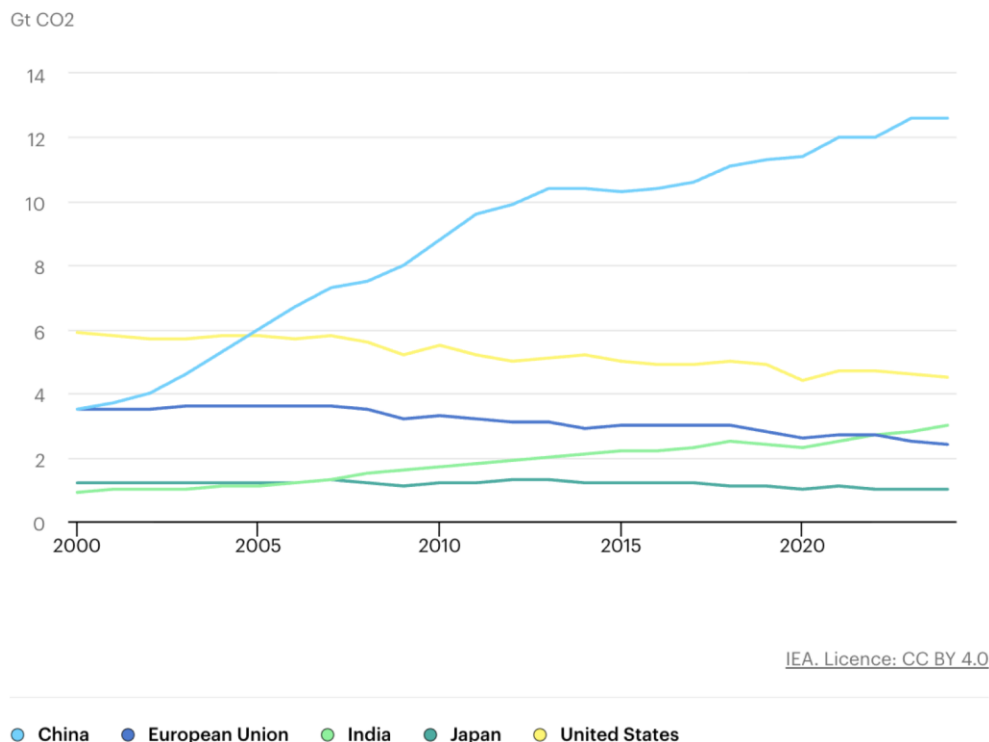


Figure 6. CO₂ Total Emissions by Region (2000-2024), (IEA 2025).

For our Methanol-to-Olefins (MTO) project, we considered the significant environmental impacts of carbon emissions. Our process begins by capturing CO₂ and converting it to methanol (MeOH) through hydrogenation. Rather than relying on conventional processes that generate additional flue gas through the combustion of natural gas or coal, our design instead utilizes steel-mill off-gases from blast furnaces, from which CO₂ is purified and recovered.

Through heat integration and process optimization, the CO₂ emissions associated with MeOH production are reduced from 3.15×10^{-1} tCO₂/tMeOH to 6.77×10^{-3} tCO₂/tMeOH, representing roughly a 98% decrease (see figure below). This reduction is supported by high process selectivity, including a CO₂ conversion of approximately 99%, enabled by an optimized CO₂:H₂ ratio (1:7) and an industrially relevant hydrogenation catalyst.

Downstream, the produced MeOH is converted into light olefins using a proprietary SAPO-34 catalyst. During this reaction, the catalyst gradually accumulates solid carbon (“coke”). To maintain catalyst performance, SAPO-34 undergoes regeneration by controlled combustion, which produces additional CO₂. In our design, this CO₂ is captured and recycled back into the MeOH synthesis loop, further lowering net emissions while also providing a modest increase in overall plant profitability.

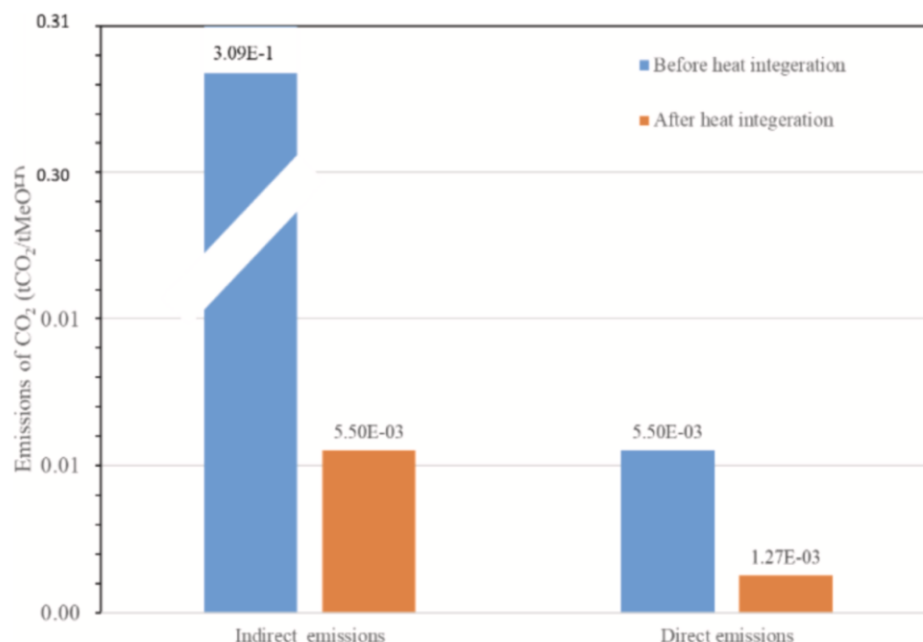


Figure 7. Total Estimated CO₂ Emissions for the Hydrogenation Process Before and After Heat Integration (Yusuf and Almomani 2023).

Our project also prioritizes efficient energy use. As mentioned earlier, the initial step of our process involves capturing CO₂ from steel-mill off-gases using an amine-based absorbent solution. Conventional amine absorption is highly energy intensive because the absorption of CO₂ is an exothermic process, while regenerating the absorbent is endothermic, requiring substantial cooling and heat input. This makes CO₂ capture one of the major operating costs in typical plants. To reduce this energy burden, our process incorporates a mixture of substituted amines that enables the absorbent solution to spontaneously form two liquid phases during CO₂ absorption: a CO₂-rich phase and a CO₂-lean phase. The CO₂-rich phase is denser, and the two phases become immiscible, allowing for separation by gravity. Only the CO₂-rich phase is sent to the regenerator, significantly lowering the required heat duty. In contrast, traditional single-phase absorbent systems send the entire absorbent stream to regeneration, leading to much higher energy consumption.

Other areas of our overall process demonstrate energy efficiency as well. In the MTO section, the fluidized-bed reactor that converts MeOH to light olefins maintains its temperature through heat supplied by the catalyst regenerator. The regenerator is directly connected to the reactor to enable catalyst recirculation, and when air reacts exothermically with the SAPO-34 catalyst during regeneration, the released heat is transferred directly into the reactor. This reduces the external energy required to maintain the optimal reactor temperature for achieving the desired selectivity and conversion.

A final example of energy efficiency in our project is found in the CO₂-to-MeOH synthesis section. Using Aspen's Energy Analyzer v11, a heat-exchanger network (HEN) is developed that significantly improved energy integration and reduced utility consumption. By strategically incorporating three heat exchangers into the final design, overall energy savings increased by 63.2%. External hot-utility demand decreased from 3.13×10^1 to 1.4×10^{-2} GW (a 99.6% reduction), and external cooling-utility demand decreased from 3.62×10^1 to 2.11×10^1 GW (a 41.7% reduction).

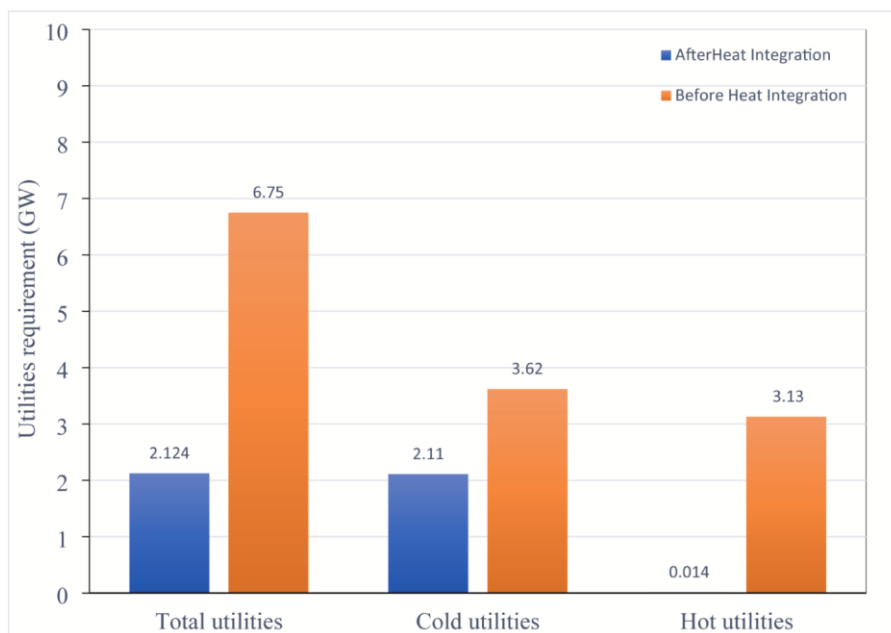


Figure 8. Required External Utility Before and After the Optimized H.E.N (Yusuf and Almomani 2023)

One additional factor that strengthens the appeal of our project is the availability of federal carbon-management incentives. Although Indiana does not impose a state-level carbon tax on industrial facilities, the federal 45Q tax credit and provisions within the Inflation Reduction Act (IRA) apply to qualifying operations. These programs offer substantial financial incentives for facilities engaged in carbon capture or utilization, including those located in Indiana. In summary, the potential for federal tax credits, the environmental advantages of our process compared to similar technologies, and the efficient use of energy collectively make our project a strong and viable endeavor.

7 Process Description

7.1 Chosen Process

This process converts carbon dioxide into high-value olefins through a fully integrated system consisting of CO₂ capture, methanol synthesis, and methanol-to-olefins (MTO) conversion. Flue gas containing CO₂ is first treated to remove solids and acidic pollutants before entering an amine absorption unit, where CO₂ is selectively captured and purified for downstream use. The CO₂ and H₂ gas are fed into a catalytic hydrogenation reactor operating over Cu/ZnO/Al₂O₃ catalyst to produce crude methanol. This mixture is cooled and flashed to remove unreacted gases before entering a purification chain, where high-purity methanol is recovered via distillation.

The purified methanol stream is then introduced into the methanol-to-olefins (MTO) reactor, where it undergoes dehydration and catalytic conversion to a mixture primarily composed of ethylene and propylene, with other byproducts such as CO₂. Downstream separation is achieved through a series of absorbers and cryogenic distillation towers. Propylene and ethylene are recovered with high purity as monomers.

In an ideal configuration, the process is expanded to create a fully integrated, carbon-circular manufacturing route to polymers. In this enhanced design, Hydrogen demand is met via on-site steam-methane reforming (SMR) or renewable-powered electrolysis. SMR-based operation includes a secondary CO₂ capture step, ensuring that nearly all carbon is retained in the system rather than emitted.

The captured CO₂ and generated hydrogen are then converted to methanol through catalytic hydrogenation in the same reactor loop used in the current design. The methanol product is dehydrated and oligomerized in an MTO reactor to form light olefins. These olefins are further processed through cryogenic separation into high-purity ethylene and propylene, just as in the current configuration. However, unlike the existing design, these monomers are not sold directly. Instead, they are immediately fed to Ziegler–Natta-based polymerization reactors to produce polyethylene and polypropylene pellets, enabling significant product value uplift and supply-chain stability.

Additionally, off-gas streams containing residual hydrogen, methane, and carbon dioxide are recycled. Hydrogen is recovered for reuse, methane is directed to reforming, and carbon dioxide is returned to the capture section, forming a closed-loop carbon cycle. Water generated during methanol synthesis and MTO conversion is treated and recycled to reduce freshwater consumption.

This idealized process configuration enhances environmental and economic outcomes by maximizing carbon utilization efficiency, minimizing purchasing of external

raw materials, increasing product value through polymer production, and enabling access to U.S. tax credits such as the federal 45Q CO₂ utilization incentive.

The following subsections provide a comprehensive explanation of the process, reaction pathways, and operating conditions involved in each major processing operation.

7.1.1 CO₂ Capture Process

7.1.1.1 Description

The selected process for the CO₂ captures that are needed to produce olefins, The patent creates a process that is designed to extract CO₂ from flue gases of large industrial plants. The source of flue gas for this project is steel plants. The process of extracting CO₂ is done through a amine separation and a distillation process. (Zhang, Qiang, Junfeng Zhang, and Jinsong Ni 2013)

The overall process has been modified from the original design in order to purify the stream for usage of steel flue gas CO₂ extraction. Flue gas to CO₂ extraction is broken down into six major processes. The first two major processes are the modification of the original design in order to use flue gas from steel plants. The patented process for Carbon capture requires a pre-treatment process for flue gas that was being processed for CO₂ separation. (Speight 2002) Therefore liberties were taken to adjust the process for processing flue gas from steel plants.

The first process is filtration to remove solids that are present in the flue gas that are in the form of dust. The dust must be removed from the flue gas before entering the next step. (Vakkilainen, Esa Kari. d) In order to remove solids, present in the flue gas, Flue gas must pass through a fiber membrane that will pick up the solids and let the rest of the gas pass through. The filter membrane is located in a bag house, but it is determined as an F-100. The dust extracted from the flue gas must be discarded as waste as the mixture of coke and metal residue and sulfur compounds are too expensive, and time consuming to separate and usually discarded when processing flue gas.

The solid free flue gas is moved into the second process that was added to a wet scrub. The process used is a ecofriendly way of safely processing flue gas in order to extract compounds that would lower the purity of the liquid CO₂ (Otoo, Enoch Kofi 2025). This step is crucial in removing compounds present in the flue gas that are similar in composition as CO₂, the patent used for the scrubbing relies on ecofriendly solvents that follow the structure of "Cat+ X-zY" to extract NO_x and SO₂ from the flue gas as the reaction inside the distillation Collom bonds with the heavier compounds leading to septation with the lighter CO₂. The simplicity of separating flue gas with ecofriendly solvents helps reduce the damage to the environment. Its reliance on this type of separation produces high quality extractions of NO_x, SO₂ that reach up to 90% purity allowing for sale of the byproducts to add into total plant revenue (Towler, Ray 2013).

The flue gas purification before reaching the original decided process are theoretical designs to apply to the concept of our future as deliberation with mentors lead to consensus of black boxing the two flue gas treatment steps to ease requirements for methanol production for olefins. (Perry, Green, and O. Maloney 1997)

To start the extraction of CO₂ from steel flue gas that has all solids and SO₂ and NO_x removed from the flue gas. The flue gas must be wet before being treasured into the amine of the column. The wetted flue gas is needed to allow ease to flow into the distillation column where a amine film is present to extract the CO₂ from the rest of the flue gas. The CO₂ binds to the amine solvent where it flows to the bottom where the CO₂ and amine are separated from each other. The amine is sent back to the the column to be reused to mitigate waste, The CO₂ follows a second distillation where water is removed from the CO₂ mixture. Once separated, the CO₂ is condensed at high pressures in order to liquify the CO₂ gas for Methanol production. The purity of the liquid CO₂ meets the expected standard 90% purity needed to be used in methanol production.

7.1.1.2 Operating Conditions

The operating conditions for extracting CO₂ from flue gas are to ensure that the CO₂ comes out in the best condition and to meet the 3,000,000 tones per year of CO₂ demand. This must be done through continuous operation throughout the year. Flue gas comes in as 22% CO₂. At proper operational conditions CO₂ purity from extraction is 90%. Conditions for equipment are for the absorber must operates at atmospheric pressure (1.00 bar) with top column temperatures of 40–50 °C, Amine solvent must be entering at 30 to 40 °C, and Amine solvent that absorbed CO₂ must be leaving at 45 to 55 °C, The stripper operates at 1.00 bar, with reboiler temperatures of about 140 °C and overhead temperatures of 60 to 95 °C, which producing 7,781.5 kmol/hr of CO₂. After condensation step, the wet CO₂ product exits at 30 °C and about 1.00 bar with a typical composition of 80 mol% CO₂ and 20 mol% water.

7.1.1.3 Alternatives

An alternative method to extract flue gas for CO₂ extraction was replacing the supply of steel mills with coal power plant flue gas. The constant emission of flue gas from burning presented a viable option due to the large number of emissions of flue gas. Using green methods to extract CO₂ from the flue gas to repurpose it in ecological friendly uses was determined to be too costly as the CO₂ purity in flue gas from coal is much contaminated with more dust than the flue gas from flue plants. Complicating the process and requirement for more solid treatment to be able to be used in olefin production.

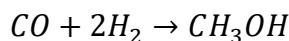
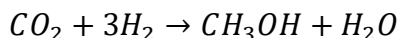
7.1.2 CO₂ to Methanol Process

7.1.2.1 Description

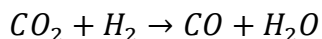
The chosen process converts carbon dioxide (CO₂) and hydrogen (H₂) into methanol (CH₃OH) through catalytic hydrogenation. Four major steps take place: Feed Preparation, Methanol Synthesis, Separation, and Purification.

The process begins by preparing and conditioning the feed streams before they enter the catalytic reactor. Carbon dioxide and hydrogen are supplied as different feeds and are initially compressed to the required reaction pressure. They are then mixed in mixing units MIX-200 and MIX-201. This combined stream forms the stoichiometric CO₂/H₂ ratio necessary for methanol synthesis. The stream is then routed through a heat exchanger that recovers heat from hot process streams. Exchanges in E-201 elevate the temperature of the feed, reducing the amount of energy required later in the heater. Finally, an external heater, H-200, raises the stream temperature to approximately 210°C, ensuring that the mixture reaches the reactor at optimal reaction conditions.

One conditioned, the feed enters the methanol synthesis reactor (R-200), a catalytic reactor packed with Cu/ZnO/Al₂O₃ catalyst. This catalyst system is widely used in industrial methanol production and is well supported by various studies (Yusuf and Almomani 2023). Within the reactor, CO₂ and H₂ undergo exothermic hydrogenation reactions to form methanol and water. The primary reactions are the following:



In addition, the reverse water-gas shift (RWGS) reaction also takes place:



Although RWGS introduces an additional reaction pathway, CO is also consumed in methanol formation, and kinetic studies have shown that CO₂ remains the main carbon source for methanol synthesis under these conditions. The reactor operates at high pressure and moderate temperature to maximize conversion, following the design conditions described by Yusuf et al. (Yusuf and Almomani 2023).

The hot reactor effluent exits R-200 and immediately enters a cooling and condensation train. Heat exchangers E-201, E-202, E-203, and E-205 remove heat from the reaction products, lowering the temperature enough to condense crude methanol and water from the gas mixture. The cooled two-phase stream flows into the first separation drum, D-200, where liquids and gases are separated. The liquid contains crude methanol, water, dissolved CO₂, and trace components, while the vapor stream consists primarily of unreacted CO₂, H₂, and small amounts of CO. To improve overall conversion, the vapor

stream is recycled back to the feed system. A small purge is taken to prevent the accumulation of inerts (Stream 14).

The crude liquid from D-200 undergoes additional cooling and flashing through exchanger E-204, and separation drum D-201 to remove remaining light gases. After gas removal, the liquid is preheated in E-205 and E-206, and directed to the purification section. Distillation column T-200 separates the methanol-water mixture into high purity methanol and water-rich bottoms stream. The overhead vapor is condensed in E-208 and collected in D-203 as the final methanol product. Overall, the process integrates compression, heat recovery, catalytic reaction, and separation into a continuous process that converts CO₂ to methanol to achieve a methanol purity of approximately 99%.

7.1.2.2 Operating Conditions

The operating conditions for the CO₂ to methanol process were selected based on published kinetic and process studies to achieve high conversion and efficient system performance (Yusuf and Almomani 2023). The CO₂ feed enters at 12.3°C and 47.63 bar, while hydrogen enters at 25°C and 30 bar. Both streams undergo compression so that the final mixed feed reaches a pressure of approximately 75-78 bar before entering the reactor. Operating at this elevated pressure is essential because the methanol synthesis reactions involve a decrease in volume, and high pressure shifts the equilibrium toward methanol formation. Before entering the reactor, the feed mixture is heated to around 210°C using a combination of heat recovery exchangers and heaters. This temperature is consistent with the optimal range for Cu/ZnO/Al₂O₃ catalysts, which show strong activity between 200 and 260°C (Yusuf and Almomani 2023).

Inside the reactor, the system operates at an inlet pressure of 75.7 bar and a controlled temperature of 210°C. These values match the specifications from the referenced study, which demonstrated that these conditions yield approximately 96% CO₂ conversion and 99% methanol production under isothermal operation (Yusuf and Almomani 2023). The reactor contains a large mass of Cu/ZnO/Al₂O₃ catalyst (44,500 kg), and a pressure drop of about 0.6 bar occurs across the reactor.

Following the reaction, the effluent enters the separation and purification stages. The first drum operates near reactor pressure (approximately 73-74 bar) to condense crude methanol without excessive vaporization. Subsequent flash drums operate at reduced pressures of 1-2 bar to remove dissolved gases and prepare the liquid mixture for distillation. The methanol purification column operates close to atmospheric pressure with a reflux ratio of 2.12. Temperatures within the column typically range between 64°C and 102°C, depending on stage composition. The condenser cools the overhead product to around 14.95°C, allowing for the collection of high purity methanol. These operating conditions allow the process to achieve stable operation, high conversion, and efficient energy use.

7.1.2.3 Alternatives

Although the original process uses the Cu/ZnO/Al₂O₃ catalyst, several alternative catalyst systems have been proposed to improve performance in CO₂ hydrogenation. The standard Cu/ZnO/Al₂O₃ catalyst offers high activity and good selectivity at moderate temperatures. However, it is sensitive to temperature fluctuations and can deactivate through copper sintering under certain conditions (Tedeewa 2024).

One potential alternative is the Cu/ZnO/ZrO₂ catalyst, where zirconia is added to increase CO₂ absorption and stabilize copper dispersion. Studies show that ZrO₂ can create stronger metal-support interactions and additional active sites, leading to improved methanol yields.

Another alternative includes oxide-based catalysts such as In₂O₃ or In₂O₃-ZrO₂. These materials activate CO₂ through oxygen vacancy sites rather than copper surfaces, which can enhance methanol selectivity and potentially reduce the influence of side reactions (Tedeewa 2024). However, indium is more expensive than copper, and large-scale validation is still needed to confirm durability and performance under typical methanol synthesis conditions.

Overall, while these alternatives show potential, they remain less employed than the Cu/ZnO/Al₂O₃ system. Given its strong industrial track record and predictable behavior under high-pressure operation, this catalyst continues to be the most practical and reliable choice for this process.

7.1.3 Methanol to Monomer Process

7.1.3.1 Description

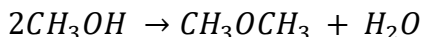
The Methanol-to-Olefins (MTO) process converts high-purity methanol into light olefins, primarily ethylene (C₂H₄) and propylene (C₃H₆), which are essential building blocks for polymers such as polyethylene and polypropylene. This process serves as an alternative to traditional olefin production from fossil feedstock, such as steam cracking of naphtha or ethane. This enables the utilization of methanol derived from sustainable or alternative carbon sources such as captured CO₂.

The MTO Reaction proceeds through an intermediate dimethyl ether (DME) formation step, followed by conversion over a shape-selective zeolite catalyst, which in this case is SAPO-34. The whole mechanism encompasses numerous intermediates, which comprise two general pathways within the hydrocarbon pool theory: the olefin cycle and the aromatic cycle. The olefin cycle is direct oligomerization and β -scission of light olefins, specifically enhancing propylene generation. The aromatic cycle is the formation of aromatic intermediates that undergo chain growth and cracking to produce both ethylene and propylene.

These reactions are strongly influenced by the catalyst's pore architecture, acidity, and diffusion constraints, which are intentionally engineered to maximize light-olefin selectivity. SAPO-34 molecular sieve catalysts are widely used in commercial MTO units because their chabazite (CHA) framework provides small cages that preferentially crack intermediates to C₂ – C₃ olefins rather than heavier hydrocarbons (Ivar M 1993),

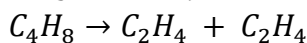
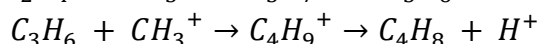
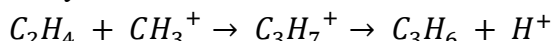
Figure 9: Chemical Reactions of MTO Process (Ivar M 1993)

1. Methanol Dehydration: formation of DME

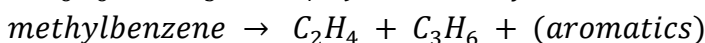
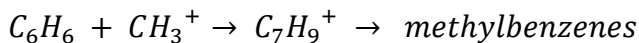
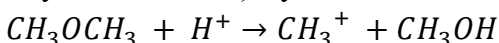


2. Hydrocarbon Pool Mechanism

- a. Olefin-Based Cycle:

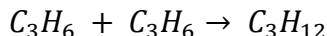


- b. Aromatic (Methylated-Benzene) Cycle:

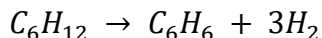


3. Side-Reactions:

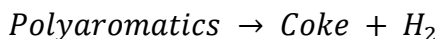
- a. Heavy Olefins Through Oligomerization



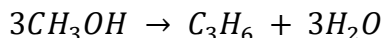
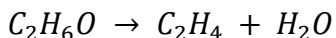
- b. Aromatics via Cyclization and Hydrogen Transfer:



- c. Coke Formation from Polyaromatics



4. Olefin Formation (Simplified Overall Reaction):



The conversion of methanol results in coke formation on the catalyst surface due to condensation and polymerization of heavier olefins. Accumulated coke reduces catalyst acidity and pore accessibility, eventually lowering olefin yield. To maintain continuous high activity, MTO systems employ a circulating fluidized-bed reactor-regenerator configuration, modeled after the fluid catalytic cracking process. In this process, fresh, hot, regenerated catalysts are continuously fed to the reactor riser and fluidized by the incoming vaporized methanol. As coke is deposited during the reaction, the spent catalyst flows to a regenerator where the coke is combusted with air. The regenerated catalyst is returned to the reactor, maintaining uninterrupted operation and a consistent activity profile.

The reactor effluent is a high-temperature mixture containing ethylene-propylene, heavy olefins and paraffins, steam and water, aromatics and oxygenates, light gases, and trace amounts of methanol and DME. To stabilize selectivity and prevent post-reactor reactions, the effluent is rapidly quenched. This, in turn, halts further conversion and protects downstream equipment. Vapor-liquid separation removes condensed water and heavier hydrocarbons for recovery or recycling, while the gas phase proceeds to fractionation.

To separate the desired products, propylene and ethylene, from the gas phase, a separation system is used. A compressor is used to ensure proper separation pressures for the water knock-out drum, which removes condensed aqueous phases. Acid-gas removal is used to eliminate CO₂ and residual catalysts. Then, the flow goes through a series of fractionation columns, including a demethanizer, deethanizer, depropanizer, and C₃ splitter. The demethanizer removes light gases, such as fuel. The deethanizer isolates the C₂ fraction. The C₂ splitter separates ethylene and ethane. The depropanizer separates the C₃ fractions. The C₃ splitter separates propylene and propane. Purified ethylene and propylene leave the system as primary salable commodities for polymerization or petrochemical processing.

MTO Technology has emerged as a key enabler, allowing process regions that lack cost competitiveness to utilize liquid petroleum feedstocks. It allows diversification of olefin production pathways, enhances energy security, and leverages carbon-recycling strategies when integrated with CO₂-based methanol synthesis.

7.1.3.2 Operating Conditions

The MTO process is operated under conditions optimized to maximize light-olefin yield while minimizing heavy hydrocarbon formation and excessive coke deposition. Prior to entering the reactor, methanol is vaporized and typically mixed with dilution steam or water to regulate hydrocarbon partial pressure and suppress secondary reactions. The vaporized feed stream is introduced to the circulating fluidized-bed reactor at temperatures typically in the range of 400–430 °C, allowing for the rapid onset of catalytic dehydration and hydrocarbon pool reactions. As the conversion proceeds exothermically, the reactor outlet temperature rises and is typically controlled within the range of 450–500 °C by adjusting the catalyst circulation rate and diluent flow.

Reactor pressure is usually maintained at slightly above atmospheric levels, generally 1–3 bar(g), to reduce compression requirements and favor the formation of light olefins. The fluidized catalyst bed ensures excellent gas–solid contact, short residence time, and uniform temperature distribution, all of which contribute to high propylene and ethylene selectivity. The ratio of steam to methanol is a crucial control parameter that varies depending on the commercial technology but typically ranges from 0.5 to 1.5 mol of steam per mol of methanol.

As coke accumulates on the catalyst during reaction, a portion of the spent catalyst is continuously withdrawn to a separate regenerator. The regenerator typically operates in the range of 650–700 °C, where controlled combustion of coke with air restores catalyst acidity and pore accessibility. Regenerator pressure is slightly higher than that of the reactor, promoting catalyst flow back into the reaction zone without relying on mechanical transport.

Downstream of the reactor, the high-temperature product stream undergoes immediate quenching to halt further cracking and stabilize olefin selectivity. The pressure at this stage remains near that of the reactor to minimize expansion losses and maintain adequate density for efficient separation. The cooled stream is then directed to multi-stage compression and fractional distillation systems, which operate at conditions tailored to achieve polymer-grade olefins, including deep refrigeration and high-pressure splitting in the ethylene and propylene purification trains.

Overall, the operating window for the MTO process is deliberately narrow and carefully controlled, as conversion, catalyst stability, and product selectivity are highly sensitive to temperature, partial pressure effects, and solids circulation rate. Effective heat management and catalyst regeneration are therefore fundamental to achieving continuous, optimal performance in commercial MTO units.

7.1.3.3 Alternatives

Some alternatives exist within the MTO process, particularly in the selection of catalysts and reactor configurations, both of which have a strong influence on olefin yield, catalyst lifetime, and overall operating efficiency.

One catalyst alternative to SAPO-34 is MAPO-18, a small pore silicoaluminophosphate material with AEI framework. MAPO-18 has slightly larger cage openings and a different acidity profile that can reduce the rate of coke formation and extend catalyst life. Studies show that MAPO-18 allows improved diffusion of intermediates and can shift the product distribution to higher propylene selectivity (Xie 2022). In addition, modifying the framework composition through metal incorporation can tune the acidity and reduce deactivation (Xie 2022), which is a major challenge for traditional SAPO-34. These properties make MAPO-18 an alternative catalyst for improving overall MTO performance.

Alongside catalyst alternatives, different reactor configurations can also be considered. Although the MTO process typically uses fluidized-bed reactors due to their ability to continuously regenerate catalyst and maintain stable temperatures, fixed-bed reactors could potentially be an alternative. Fixed-bed reactors are simpler to construct and have lower capital costs. However, they face challenges such as coke deposition and faster catalyst deactivation compared to fluidized systems (Yuan 2017).

Overall, these alternatives demonstrate some flexibility within the MTO process. However, each alternative comes with trade-offs in catalyst lifetime, regeneration needs, and overall efficiency. Therefore, further research is needed to determine the combination of catalyst and reactor design that provides the best balance of performance and economic feasibility for the MTO process.

7.2 Competing Processes

7.2.1 Methanol to Gasoline

7.2.1.1 Description

The Methanol-to-Gasoline (MTG) process is a catalytic conversion method that produces high-octane gasoline from methanol. The process begins by vaporizing and pre-heating crude methanol, which is sent to a dimethyl ether (DME) reactor. In this reactor, ethanol is partially dehydrated over alumina catalyst to form a mixture of DME, methanol, and water. This reaction is slightly exothermic and reaches an equilibrium mixture at moderate temperatures.

Next, the DME-rich stream flows into the ZSM-5 catalyst reactors, where the major conversion to hydrocarbons occurs. ZSM-5 promotes a complex series of reactions that rearrange methanol and DME molecules into gasoline hydrocarbons. These products include normal paraffins, isoparaffins, aromatics, and small amounts of light gases. Most of the reaction heat is released in this step.

The reactor effluent is then cooled and separated into gas, water, and hydrocarbon liquid. The gas stream (mostly hydrocarbons, CO and CO₂) is partially recycled to moderate reactor temperature and maintain stable operation. The water phase, which contains small, oxygenated compounds, is sent to biological treatment.

The hydrocarbon liquid, known as raw MTG gasoline, is distilled to remove light gases and undesirable components. One important treatment step is the heavy gasoline treating unit (HGT), which removes durene (1,2,4,5-tetramethylbenzene). After durene reduction, the treated gasoline is blended with other fractions to meet product specifications. A block flow diagram of the process is shown below.

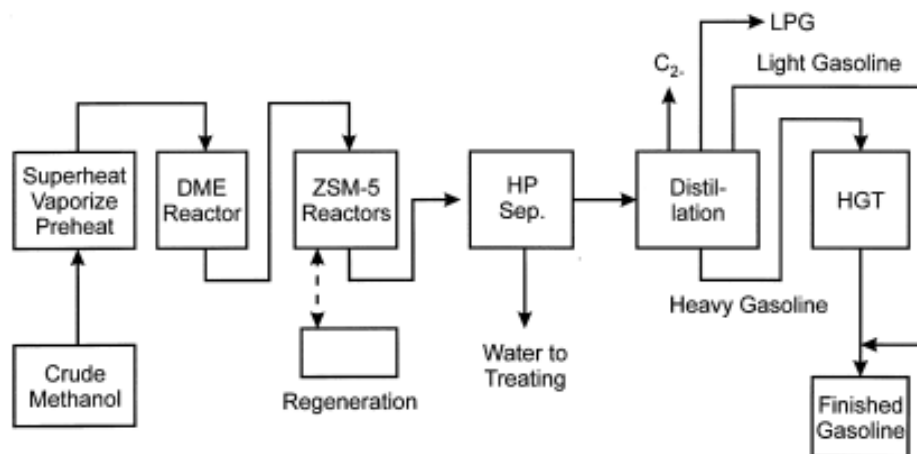


Figure 10: MTG Block Flow Diagram (Keil, 1999).

Like any industrial process, MTG has both advantages and disadvantages. One major advantage is that the gasoline produced has high octane quality and can be used directly in existing engines. The process is also appealing because it can be made from non-petroleum derived sources, giving it potential environmental value. In contrast, some disadvantages relate to catalyst performance and product composition. The ZSM-5 catalyst gradually deactivates due to coke build-up, requiring periodic regeneration. Additionally, the process naturally forms a significant amount of durene, which must be removed to ensure acceptable fuel quality. The MTG reaction is also highly exothermic, so careful temperature management is required. Despite these challenges, the MTG process remains a proven route for producing gasoline from methanol.

7.2.1.2 Operating Conditions

The MTG process operates under well-defined conditions to ensure efficient conversion and stable catalyst performance. In the first stage, the DME reactor typically runs at temperatures of about 310-320°C and a pressure close to 26 bar. These conditions allow methanol to reach its equilibrium mixture with dimethyl ether and water while releasing a modest portion of the overall process heat.

The second stage operates at higher temperatures of approximately 350-366°C and pressures of 19-23 bar. The ZSM-5 reactors are where most of the reaction heat is released, as the DME and methanol molecules are rearranged into gasoline hydrocarbons. To maintain stable operation, the reactors are typically arranged in parallel, with one reactor periodically being deactivated for catalyst regeneration. As the effluent exits the reactors, it is cooled to 25-35°C using cooling water before entering a three-phase separator, where gas, water, and liquid hydrocarbons are removed. Downstream distillation and treating steps then separate the gasoline fractions and remove unwanted components such as durene. These controlled operating conditions allow the MTG

process to achieve high methanol conversion, consistent gasoline quality, and reliable long-term performance.

7.2.2 Ethanol to Ethylene

7.2.2.1 Description

The process converts ethanol into ethylene using a fluidized-bed dehydration reactor with catalyst regeneration. Ethanol vapor enters through the bottom of the fluidized-bed reactor, where it interacts with a catalyst for such alumina. The upward flow of ethanol maintains the catalyst in a fluidized state, ensuring excellent heat transfer and uniform temperature. Inside this reactor, ethanol is dehydrated into ethylene at high temperatures around 370 to 480 Celsius. This allows for very short processing times that last around 10 seconds to complete. The short processing time allows for a very high yield of 90% conversion.

The reactor waste product consists of ethylene and water; the waste products can be easily repeated downstream in the olefin production step. A portion of the catalyst continuously circulates out of the reactor as a way to mitigate the use of the new catalyst for lower cost and waste emissions. A regeneration step can be used to remove water from the spent catalyst. The regenerator consists of two streams used to pass hot air for drying and a vent and a second stream to push out humid air.

The high temperature from the streams that supply the reactor can be combined with cooler streams that require heating such as stream 2 and stream 4. To mitigate the use of energy in heating according to the patent concept of ethane to ethylene processing. Reactor flue gases can be cycled into the initial filtration step to decrease waste and increase CO₂ extraction in order to lower waste production exit through a vent line.

Overall, the system must operate continuously as the constant feed of ethanol leads to a constant amount of waste products that can be reincorporated into the overall system as a viable alternative to olefin production. The issue of running this with other parts of the olefin and CO₂ production is with the amount of heat used to produce olefins required is the amount of heat needed to run an ethylene reaction is too high and must be contained.

7.2.2.2 Operating Conditions

To operate this plant, the ethanol dehydration reactor must operate at 345 to 480 °C. With vaporized ethanol contacting a fluidized bed of silica–alumina catalyst and a very short residence time of 1 to 10 seconds. The endothermic reaction is supplied heat by circulating regenerated catalyst from a separate regenerator, which operates at 1540 to 705 °C with preheated air to burn off coke and restore catalyst activity. The regenerated catalyst returns hotter than the reactor to sustain reactor temperature, while flue gas

leaves the regenerator slightly above atmospheric pressure. Overall ethanol exceeds 99%, while ethylene yields typically 95%.

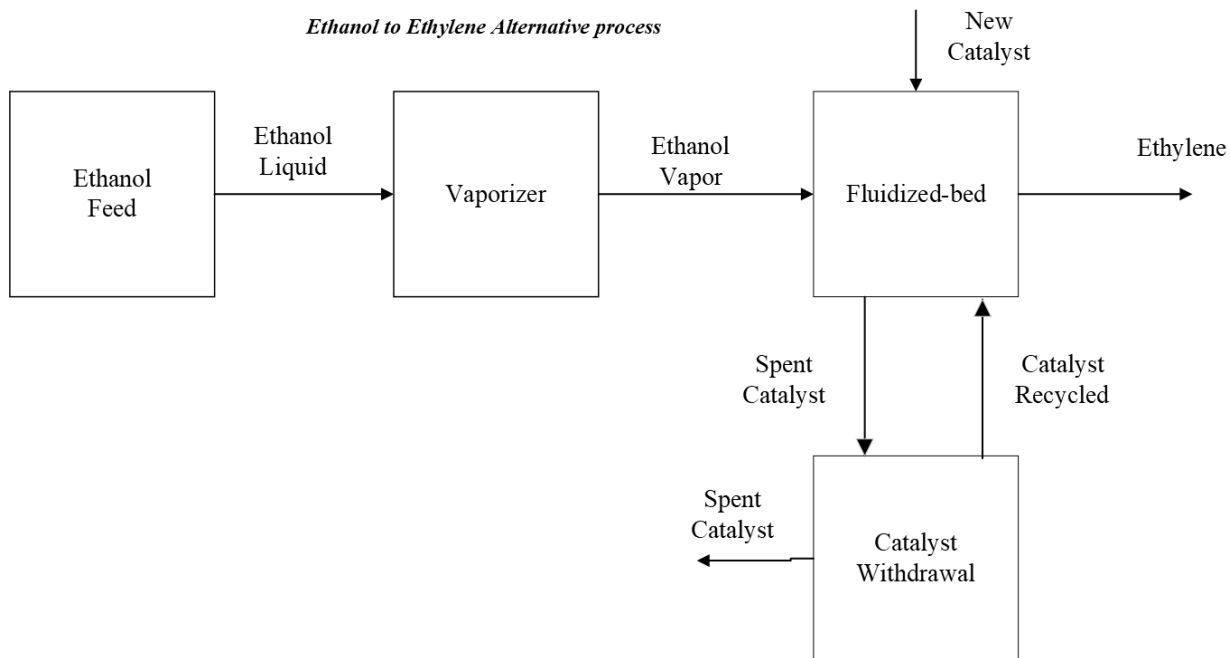


Figure 11: Ethanol to Ethylene Block Flow Diagram

7.2.3 CO₂ to Syngas Process

Due to the harmful effects of CO₂, many alternative projects exist that capture and utilize CO₂ in different ways. One alternative process involves converting CO₂ to synthesis gas (or syn gas) via RWGS using methane as a hydrogen and carbon source that's obtained using steam methane reforming (SMR). The heat source for the SMR and RWGS reactors is generated through electrical energy, eliminating the need for hydrocarbon fuels.

The conversion of carbon dioxide to syngas through the reverse water-gas shift (RWGS) reaction is a multi-stage process that integrates chemical transformations with essential unit operations to achieve efficient CO₂ utilization. The process begins with CO₂ feed preparation, in which captured carbon dioxide, typically sourced from flue gas or direct air capture, is purified and compressed to the required operating conditions. In a parallel process, hydrogen production is carried out via water electrolysis, typically powered by renewable electricity, ensuring a sustainable supply. These two streams are introduced into the RWGS reactor, an electrified endothermal unit maintained at high temperatures (700–900°C) through resistive or “Joule” heating. Within this reactor, the primary reaction is the reverse water-gas shift, where CO₂ and H₂ react to form CO and H₂O. Secondary reactions may occur depending on operating conditions, including the forward water-gas shift ($\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$), which can partially equilibrate the

system, and methanation side reactions ($\text{CO} + 3\text{H}_2 \rightarrow \text{CH}_4 + \text{H}_2\text{O}$), though these are minimized under RWGS conditions to preserve syngas selectivity.

Following the reactor stage, the process employs several unit operations to condition and separate products. Heat exchangers recover sensible heat from the hot effluent stream, improving overall energy efficiency. Condensers remove water vapor formed during the RWGS reaction, leaving a dry syngas mixture. Gas separators or membrane units may be used to adjust the H_2/CO ratio to meet downstream synthesis requirements, such as methanol or Fischer–Tropsch fuel production. In some configurations, acid gas removal (AGR) units are integrated to eliminate residual CO_2 , ensuring a clean syngas stream. The final product is a tailored mixture of CO and H_2 , ready for subsequent chemical synthesis.

This process flow demonstrates the interplay between chemical reactions and engineering operations. The RWGS reaction provides the fundamental transformation of CO_2 into CO , while unit operations such as compression, heating, condensation, and separation ensure that the syngas is produced efficiently, selectively, and in a form suitable for industrial application. The electrification of the reactor distinguishes this pathway from conventional thermal systems, offering precise temperature control, rapid response to fluctuating energy inputs, and compatibility with renewable electricity. Together, these features position the electrified RWGS process as a technically robust and environmentally favorable route for CO_2 utilization, contingent on continued advancements in catalyst stability, reactor design, and integration with renewable hydrogen supply chains.

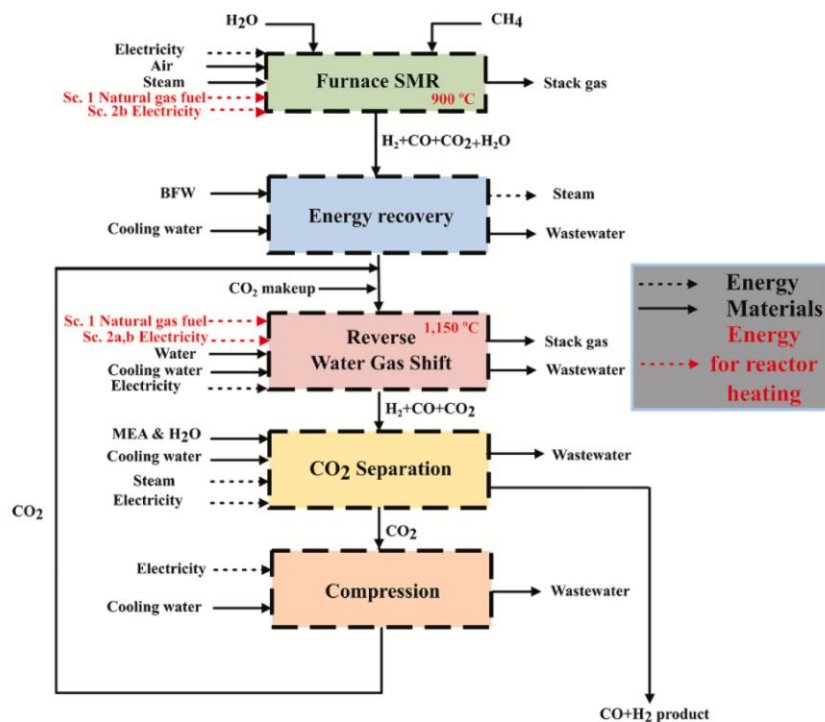


Figure 12. SMR + RWGS for CO₂ to Syngas (Cao, 2022).

8 Conclusion and Next Steps

This project shows how captured CO₂ from a steel plant's flue gas can be converted into useful polymer products. By converting CO₂ into methanol and then into olefins, the process helps reduce industrial carbon emissions while also meeting the growing demand for plastics. Light olefins, such as ethylene and propylene, are used for the production of many everyday materials, and finding cleaner ways to produce them is becoming increasingly important as industries look for lower-carbon alternatives to traditional fossil-based routes.

However, producing only ethylene and propylene is not enough to maximize the value of this process. The market prices for polyethylene (PE) and polypropylene (PP) are significantly higher than the prices of their monomers. Because the suggested process requires a large amount of hydrogen, which has a high cost, it is necessary to create higher value final products to make the project financially feasible. Selling the monomers directly would not generate enough profit to balance out the high hydrogen costs, for which adding polymerization to the process is an essential step. By turning ethylene into PE and propylene into PP, the plant can sell products with much higher market value, which greatly improves the economic outlook of the entire system.

Even though this design can significantly reduce CO₂ emissions, further improvements are needed to make it commercially competitive. One of the biggest challenges is the cost of hydrogen. Future work should focus on finding ways to lower hydrogen production costs by integrating steam methane reforming (SMR) with CO₂ capture. Additional ways to reduce overall costs include improving heat integration to reduce energy consumption, optimizing reactor and separation performance, reducing the number of compressors being used, and extending catalyst life to minimize replacement costs.

Overall, this project shows that converting CO₂ into polymers would be a promising approach to lower carbon emissions in plastic manufacturing. With improvements that reduce production costs and increase overall efficiency, especially in hydrogen supply and energy use, the process has the potential to become both environmentally friendly and economically sustainable.

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10 Appendix

10.1 Design Basis

The objective of design is to utilize post-combustion CO₂ emissions from steel refinery operation as a sustainable carbon feedstock for the production of polymer-grade light olefins, specifically ethylene and propylene. The process integrates CO₂ capture =, hydrogen supply, methanol synthesis, and methanol-to-olefin (MTO) conversion in a continuous production facility with high energy efficiency and minimal environmental impact.

10.1.1 Plant Size

The proposed facility is designed for large-scale olefin production to meet industrial demand while achieving significant CO₂ reuse.

<i>Table 3: Plant Size Specifications</i>	
Parameter	Value
Annual Operating Time	8000 h/y
Construction Period	2 years
Start-up Capacity	50% design capacity during 3 rd year
Plant Life	20 years
Total Light Olefins Production	830,000 t/ year
Ethylene Production	360,869 t/year (polymer grade)
Propylene Production	469,130 t/year

10.1.2 Raw Material Requirements

The raw material supply is based on the existing steel refinery exhaust streams supplemented by necessary reaction inputs.

The amount of flue gas needed to operate plant is 7.32×10^9 m³/yr., which contains CO₂, N₂, O₂, H₂O, Solids, NO_x, SO₂ and CO.

<i>Table 4: Composition of Flue Gas</i>	
Component	Composition (mol%)
CO ₂	23.0 mol%
N ₂	51.0 mol%
O ₂	5.0 mol%
H ₂ O	1.0 mol%
Solids	0.5 mol%
NO _x	6.0 mol%
SO ₂	3.0 mol%
CO	4.5 mol%

The additional process inputs include water, steam, hydrogen gas, oxygen, and nitrogen. The consumables include the Cu/ZnO/Al₂O₃, SAPO-34, amine solvent, and nitrogen.

10.1.3 Location

The location for this process, Gary, Indiana, was chosen based on the proximity to steel mills, existing industrial infrastructure, and transportation options for product distribution. Furthermore, the site characteristics, such as the low elevation, low ambient pressure, and maximum and minimum temperatures allow for good conditions for this plant. Most importantly, the regional availability of industrial utilities and the large volume of nearby CO₂ emissions make Gary, IN an ideal location for circular carbon technology facility.

10.2 Block Flow Diagram

10.2.1 Current Process

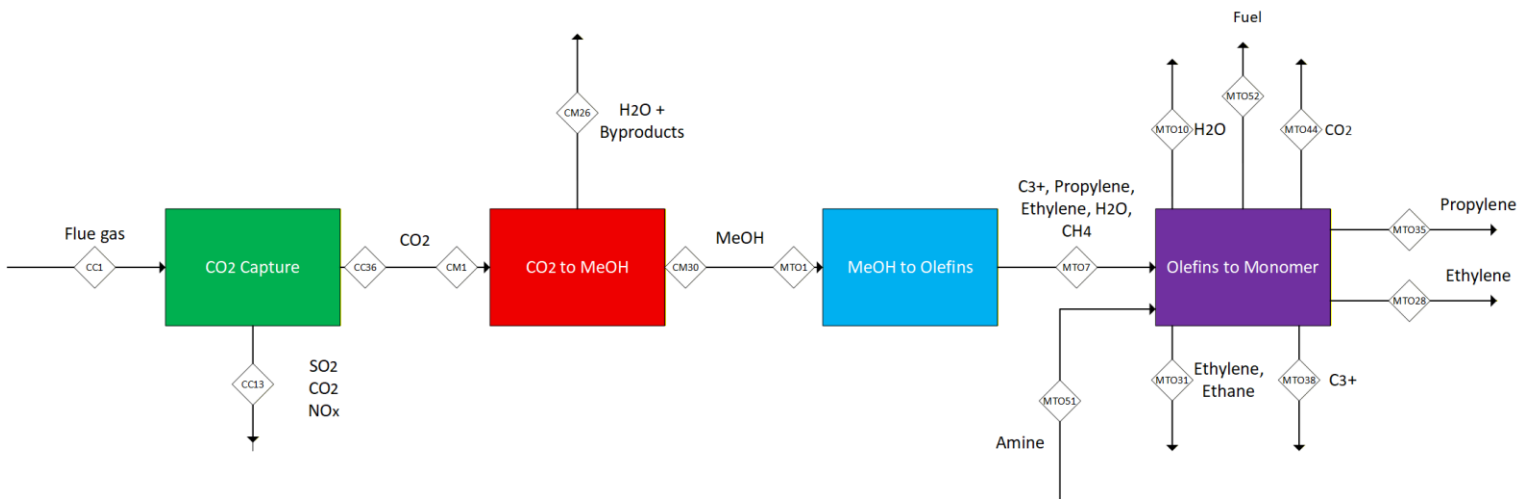


Figure 13: BFD of the Current Process

10.2.2 Ideal Case Process

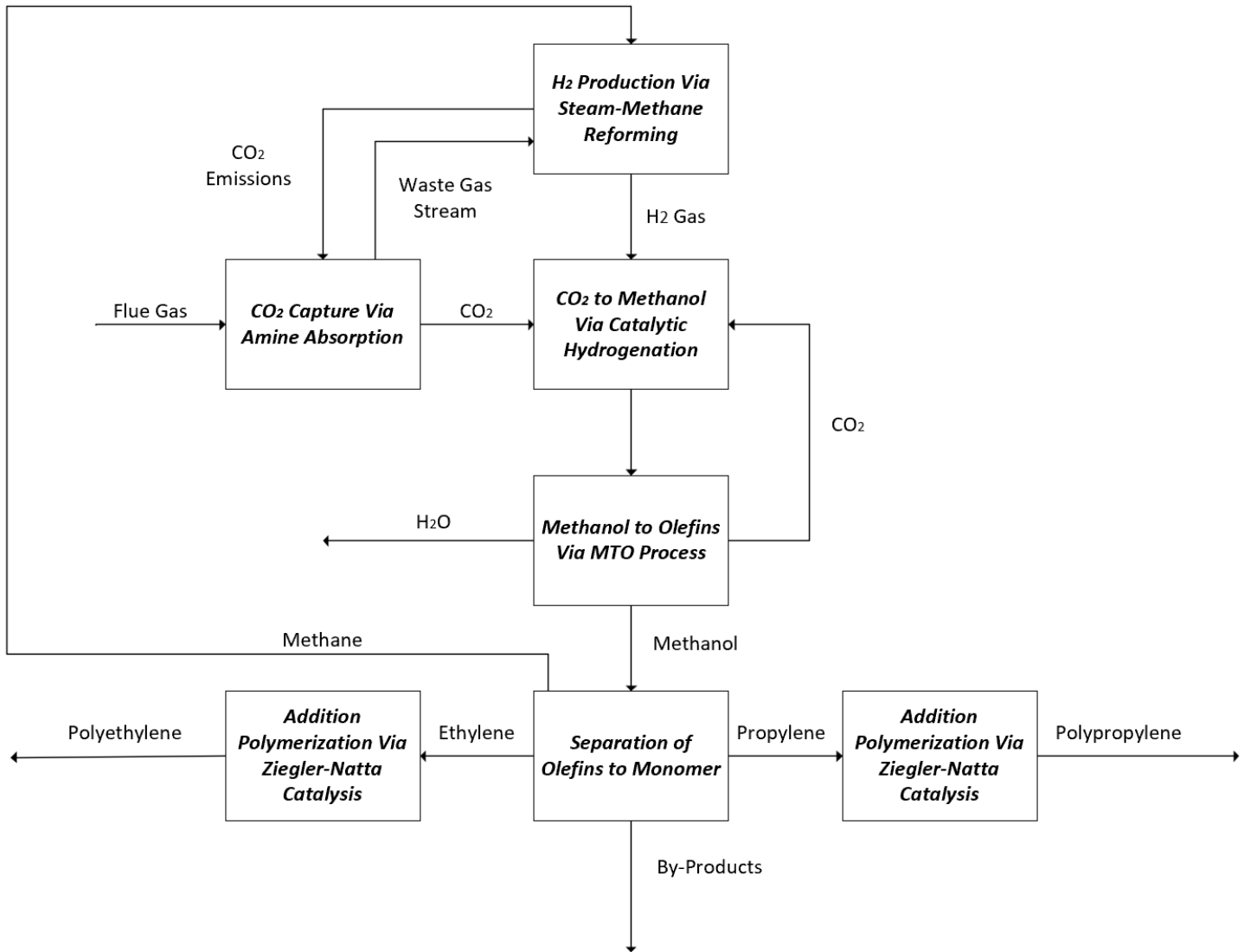


Figure 14: Block Flow Diagram of the Proposed Ideal Process

10.3.1 CO₂ Capture Process

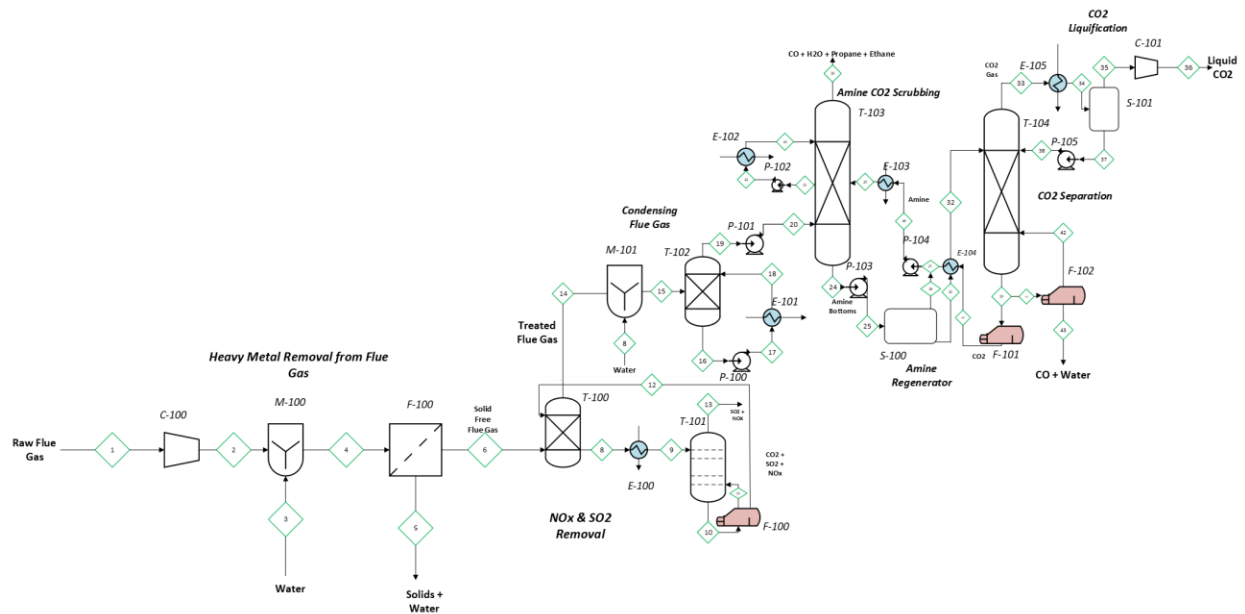


Figure 15: CO₂ Processing from Flue Gas with Omitted Pretreatment steps of Flue Gas

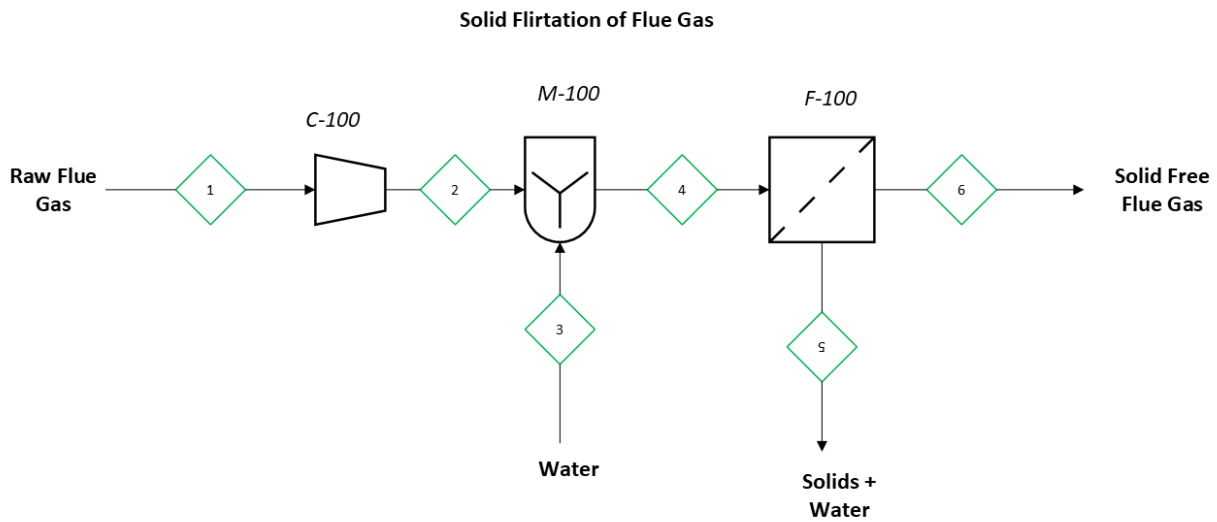


Figure 16: CO₂ Process of Removing Solids from Flue Gas

10.3.2 CO₂ to Methanol Process

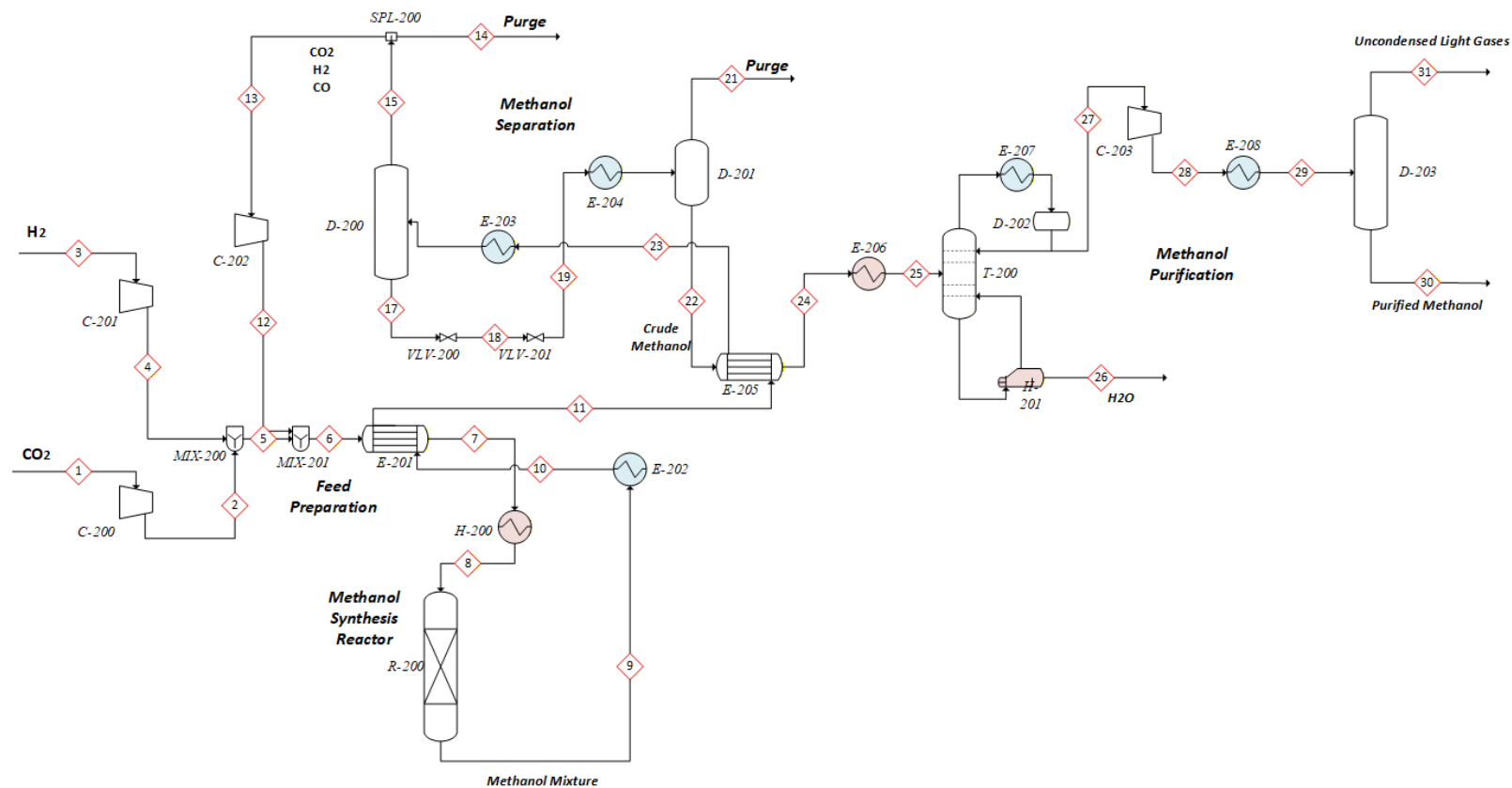


Figure 19: CO₂ to Methanol Process Flow Diagram

10.3.3Methanol to Monomer Process

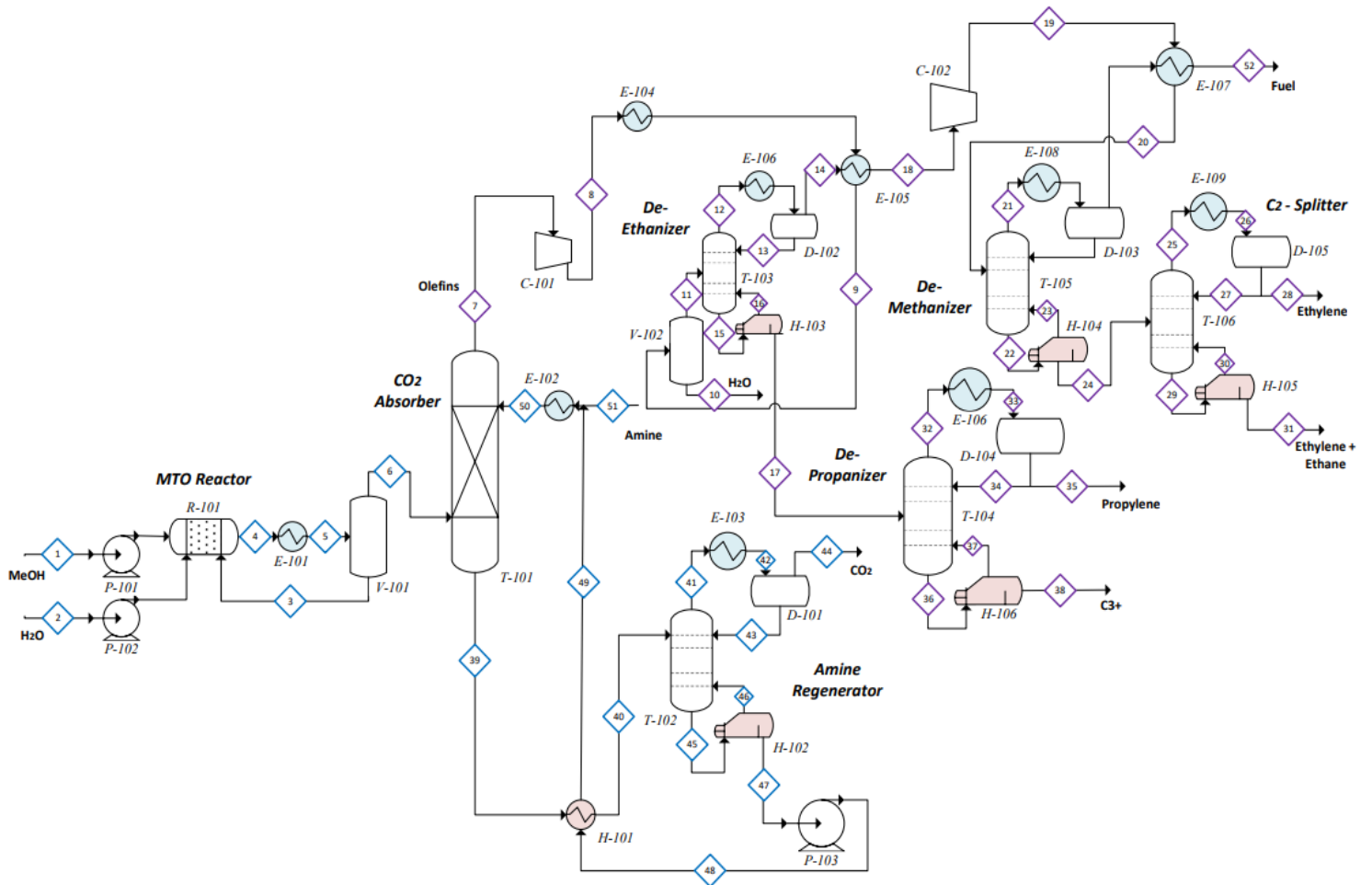


Figure 20: Methanol to Monomer Process Flow Diagram

10.4 Rough Material Balance

10.4.1 CO₂ Capture Process

<i>Table 5: Rough Material Balance of Methanol to Monomer Process</i>										
Stream	Flue Gas	Water	Coke	Iron Dusts	CO	N ₂	SO ₂	NO _x	CO ₂	Amine
1	1									
2		1								
3		0.6	0.2	0.2						
4	0.9	0.1			0.1	0.48			0.22	
5		1								
6		0.2					0.4	0.4		
7		0.3			0.3	0.2			0.2	
8		0.3			0.3	0.2			0.2	
9		0.3			0.3	0.2			0.2	
10		0.3			0.3	0.2			0.2	
11		0.3			0.3	0.2			0.2	
12		0.3			0.3	0.2			0.2	
13		0.3			0.3	0.2			0.2	
14		0.3			0.3	0.2			0.2	
15		0.2			0.3	0.3			0.1	0.1
16		0.2			0.3	0.3			0.1	0.1
17		0.2			0.3	0.3			0.1	0.1
18		0.2							0.3	0.5
19		0.2							0.3	0.5
20		0.2							0.3	0.5
21		0.2							0.3	0.5
22		0.2							0.3	0.5
23		0.2							0.3	0.5
24		0.3			0.1	0.3	0.3			
25									0.7	0.3
31		0.5			0.1				0.2	0.2
32		0.6			0.4					
33										
34		0.1			0.1				0.8	
35		0.1			0.1				0.8	
36		0.1			0.1				0.8	

37		0.1							0.9	
38		0.1							0.9	
39		0.1							0.9	

10.4.2 CO₂ to Methanol Process

Table 6: Rough Material Balance of CO₂ to Methanol Process

Stream	CO ₂	H ₂	MeOH	Water	CO	N ₂
1	1					
2	1					
3		1				
4		1				
5	0.74	0.260				
6	0.74	0.260				
7	0.74	0.239				0.021
8	0.74	0.239				0.021
9	0.028	0.051	0.508	0.343	0.071	
10	0.028	0.051	0.508	0.343	0.071	
11	0.028	0.051	0.508	0.343	0.071	
12	0.207	0.133			0.396	0.264
13	0.207	0.133			0.396	0.264
14	0.207	0.133			0.396	0.264
15	0.207	0.133			0.396	0.264
16	0.207	0.133			0.396	0.264
17	0.207	0.133			0.396	0.264
18	0.207	0.133			0.396	0.264
19			0.569	0.431		
20			0.569	0.431		
21	0.207	0.133			0.396	0.264
22			0.569	0.431		
23			0.569	0.431		
24			0.569	0.431		
25			0.569	0.431		
26			0.001	0.999		
27	0.057	0.066			0.42	0.457
28	0.057	0.066			0.42	0.457
29			0.569	0.431		

30			0.999	0.001		
31	0.057	0.066			0.42	0.457

10.4.3 Methanol to Monomer Process

Table 7: Rough Material Balance of Methanol to Monomer Process

Stream	MeOH	Water	C2H4	C3H6	Fuel	C ₃ ⁺	CO2	Amine	C2H6
1	1.00								
2		1.00							
3	0.64	0.36							
4	0.10	0.10	0.20	0.20	0.10	0.10	0.10		0.10
5	0.10	0.10	0.20	0.20	0.10	0.10	0.10		0.10
6		0.05	0.30	0.25	0.10	0.10	0.10		0.10
7		0.10	0.30	0.30	0.10	0.10			0.10
8		0.10	0.30	0.30	0.10	0.10			0.10
9		0.10	0.30	0.30	0.10	0.10			0.10
10		1.00							
11			0.35	0.30	0.13	0.12			0.10
14			0.50		0.20				0.30
17				0.80		0.20			
18			0.50		0.20				0.30
19			0.50		0.20				0.30
20			0.50		0.20				0.30
24			0.90						0.10
28			1.00						
31			0.20						0.80
35				1.00					
38						1.00			
39							0.30	0.70	
40							0.30	0.70	
44							1.00		
47								1.00	
48								1.00	
49								1.00	
50								1.00	
51								1.00	
52					1.00				